

Python for Data Science (PDS)

Lab - 15

Roll No. : 128

Name : Riyaz

Given an image file:

```
In [3]: from PIL import Image #Python Imaging Library (PIL)
```

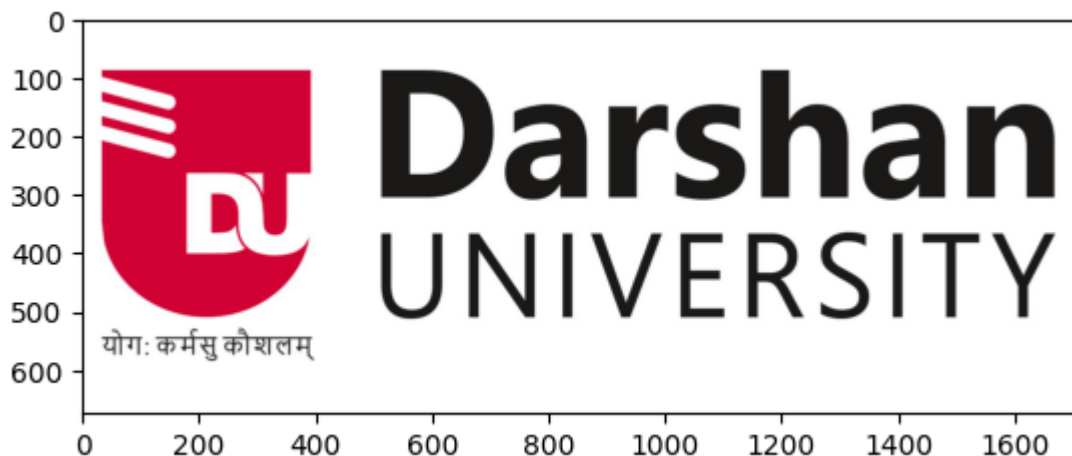
```
In [5]: import numpy as np
```

```
In [7]: import matplotlib.pyplot as plt
```

01) Read the image using the PIL library.

```
In [10]: img1 = Image.open("logo.png")
```

```
In [12]: plt.imshow(img1)  
plt.show()
```



```
In [14]: plt.imshow(img1)  
plt.axis("off")  
plt.show()
```



योग: कर्मसु कौशलम्

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02) Convert the image into a NumPy array.

```
In [17]: img1_array = np.array(img1)
```

```
In [19]: print(img1_array.shape)
```

```
(675, 1705, 3)
```

03) Create a copy of the image array.

```
In [22]: img_copy = img1_array.copy()
```

04) Extract a sub-image starting from the 100th row and 100th column, selecting every 20th row and column.

```
In [25]: subimg = img_copy[100::20, 100::20]
```

05) Display the processed image using Matplotlib.

```
In [28]: plt.imshow(subimg)  
plt.axis("off")  
plt.show()
```



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